

The Chain Lemma

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Statement. Fix integers $m \geq 2$ and r with $0 \leq r \leq m-1$, and let c be any residue with $0 \leq c \leq m-1$ and $c \neq r$. In $D(m, r)$, for every position $n \geq 0$ with $n \equiv c \pmod{m}$:

n is a P-position $\iff n \equiv c \pmod{2m}$.

Setup and notation. Chain c is the set $\{c + jm : j = 0, 1, 2, \dots\}$, i.e. all positions $\equiv c \pmod{m}$; call $c + jm$ rung j . Two arithmetic remarks used throughout. (i) Every rung satisfies $c + jm \equiv c \not\equiv r \pmod{m}$, so every position of the chain is in fallback mode: its only candidate move is subtract- m , legal exactly when the position is $\geq m$. (ii) Since $n \equiv c \pmod{m}$, the residue of $n \bmod 2m$ is either c or $c + m$; and $n = c + jm \equiv c \pmod{2m} \iff 2m \text{ divides } jm \iff j \text{ is even}$. So the Lemma is equivalent to: rung j is P iff j is even. The definition doing all the work: a terminal position is P; a position is N iff at least one legal move reaches a P-position; a position is P iff every legal move reaches an N-position, the terminal case being this clause satisfied vacuously.

Closure. Let $n = c + jm$ be a rung. Its only candidate move is $n \rightarrow n - m$. If $j = 0$, the move is illegal, since $n = c \leq m - 1 < m$. If $j \geq 1$, the move is legal, since $n = c + jm \geq 0 + m = m$, and it lands on $c + (j-1)m$: the same residue $c \bmod m$, hence rung $j - 1$ of the same chain. So every legal move from a chain- c position stays inside chain c , descending exactly one rung, and the set of positions reachable from the chain is contained in the chain.

Independence. Positions outside chain c — in particular class- r positions firing squares — may well have moves that land on rungs of the chain. This cannot disturb the rungs' statuses: by the definition just quoted, the status of a position is a function only of the statuses of its options, the positions reachable from it; the definition never mentions moves arriving at a position. By Closure, every option of every rung is again a rung, so the statuses of chain c are computed entirely within chain c and would be identical if the rest of $D(m, r)$ were deleted. Incoming traffic affects only the travellers, never the rungs.

The bottom. Rung 0 is the position c , the least element of the chain, with $0 \leq c \leq m-1$. It is in fallback mode (because $c \not\equiv r$) and its single candidate move is illegal (because $c < m$). Both facts hold equally for $c = 0$ and for $c \geq 1$ — note that for $c = 0$ no appeal to squares is needed: $0 \neq r$ already puts the position 0 in fallback mode, where $m \leq 0$ fails. So rung 0 is terminal, hence a P-position: “every legal move reaches an N” holds vacuously. This matches the claim, since $j = 0$ is even.

Alternation. Claim: rung j is P for even j and N for odd j . Strong induction on j . Base, $j = 0$: done above. Step, $j \geq 1$: assume the claim for all smaller indices. By Closure, rung j has exactly one legal move — legal precisely because $j \geq 1$, the boundary check (rung 0 has no move at all; rung 1's move is legal since $c + m \geq m$) — and it goes to rung $j - 1$. For a position with a single option x , the definition specializes: “some option is P” means x is P, so the position is N iff x is P; “every option is N” means x is N, so the position is P iff x is N. By the induction hypothesis, rung $j - 1$ is P iff $j - 1$ is even. Hence rung j is N iff $j - 1$ is even, i.e. iff j is odd; and rung j is P iff $j - 1$ is odd, i.e. iff j is even. Induction closed.

Assembly. Let $c \neq r$ be any class and $n \equiv c \pmod{m}$ any position; write $n = c + jm$ with $j \geq 0$ uniquely determined. Closure and Independence license computing n 's status inside the chain alone; the bottom is the base of the induction and Alternation is its step, giving: n is P $\iff j$ is even $\iff jm \equiv 0 \pmod{2m} \iff n \equiv c \pmod{2m}$. Nothing about c was used beyond $0 \leq c \leq m-1$ and $c \neq r$, so the result holds for all $m - 1$ such classes simultaneously. ■

Scope. The Lemma classifies no position of class r itself — it is silent about every $n \equiv r \pmod{m}$, and in particular does not assert those are N-positions; that claim is Milestone 2, and only the two together give the full mod- $2m$ law.

Closing remarks: checks and observations

Workbook, $D(5,2)$, $n = 0 \dots 24$, from the raw definition: P at 0, 1, 3, 4, 10, 11, 13, 14, 20, 21, 23, 24; N everywhere else — exactly residues 0, 1, 3, 4 $\pmod{10}$, with class 2 ($n \equiv 2, 7 \pmod{10}$) all N so far. The decoding in the brief checks out. Also verified mechanically to $n = 400$: no violations of the Lemma in $(5,2)$, $(3,1)$, $(7,3)$, $(2,1)$, or $(4,1)$ for any class $c \neq r$.

The one honest observation about the diseased pages: the chains stay healthy even where the game is sick. In $D(2,1)$ and $D(4,1)$ every non- r chain alternates exactly as the Lemma predicts; what breaks is class r , which grows P -positions of its own — first at $n = 3$ in $(2,1)$ and $n = 5$ in $(4,1)$ (then 11, 23, ... and 13, 37, ...). So the disease lives entirely in Milestone 2, not in this page — consistent with the fact that the proof above never once mentions squares.